ASSOCIATED DETAIL SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
NAND BUFFERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced Schottky TTL NAND buffers. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete number part.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Quad 2-input NAND buffer
02	Quad 2-input NAND buffer (open collector)
03	Dual 4-input NAND buffer

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device type 01	182 mW
Device type 02	165 mW
Device type 03	94 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases C and D	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 4/:	
Low level 01, 03	80 maximum
Low level 02	16 maximum
High level 01, 03	50 maximum
High level 02	12 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

4/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	01, 03	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 48 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	ALL		0.55	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_L = -18 mA T_C = +25°C	ALL		-1.2	V
High level input current	I_{IH1} I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V V_{CC} = 5.5 V, V_{IN} = 7.0 V	ALL		20 100	µA µA
Low level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IL} = 0.5 V	ALL	0	-1.2	mA
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V, V_{OS} = 0.0 V	01, 03	-100	-225	mA
Maximum collector cut-off current	I_{CEX}	V_{CC} = 4.5 V, V_{IN} = 0.8 V	02		250	µA
Supply current	I_{CCL} I_{CCH}	V_{CC} = 5.5 V	01 01		33 6	mA mA
Supply current	I_{CCL} I_{CCH}	V_{CC} = 5.5 V	02 02		30 7	mA mA
Supply current	I_{CCL} I_{CCH}	V_{CC} = 5.5 V	03 03		17 4	mA mA
Propagation delay time high to low level	t_{PHL}	V_{CC} = 5.0 V R_L = see figure 3 C_L = 50 pF	01	1.0	5.5	ns
Propagation delay time high to low level	t_{PHL}		02	1.0	6.0	ns
Propagation delay time high to low level	t_{PHL}		03	1.0	5.5	ns
Propagation delay time low to high level	t_{PLH}		01	1.5	7.0	ns
Propagation delay time low to high level	t_{PLH}		02	6.5	14.5	ns
Propagation delay time low to high level	t_{PLH}		03	1.0	6.5	ns

1/ Not more than one output should be shorted at a time.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9, 10, 11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3, 9,10,11	1,2,3,9
Group B test requirements (method 5005) subgroup 5	1,2,3, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, E, or F using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

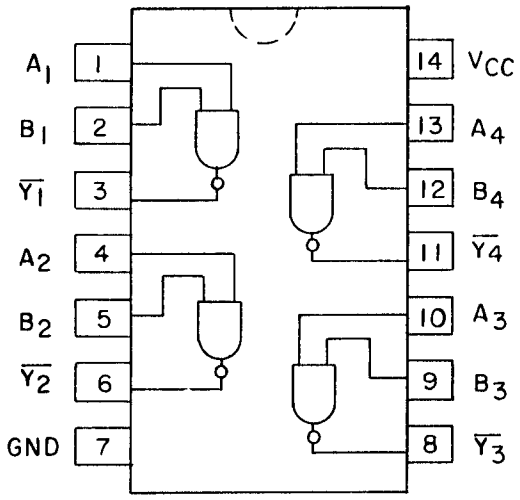
b. Interim and final electrical test parameters shall be as specified in table II, herein except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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Device type 01, 02

Case C and D



Device type 01, 02

Case 2

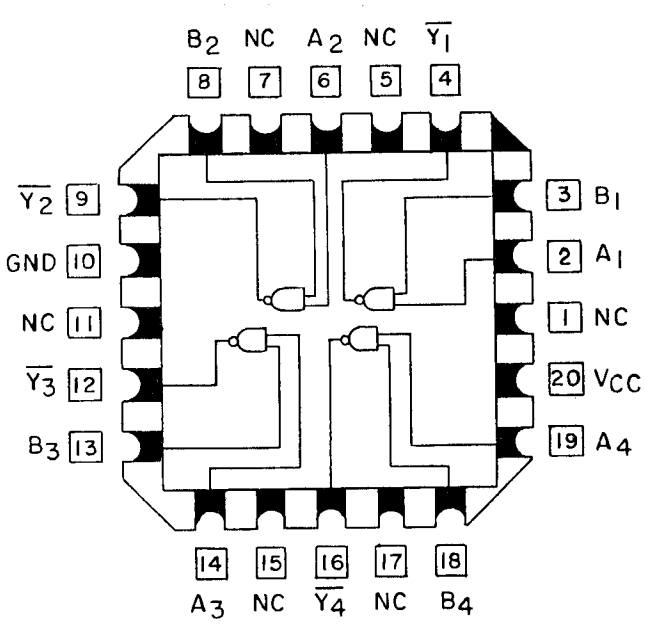
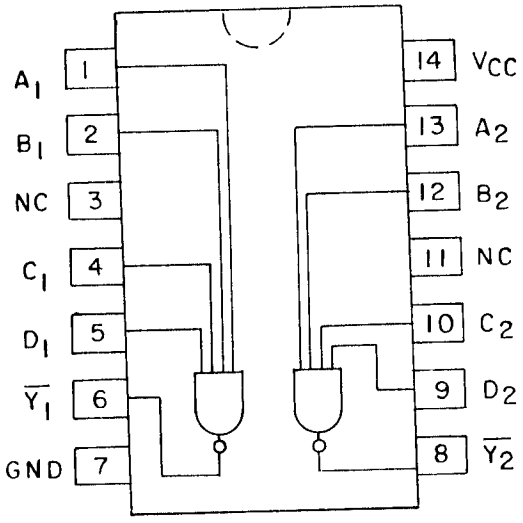


FIGURE 1. Terminal connections and logic diagram (top view).

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Device type 03

Cases C and D



Device type 03

Cases 2

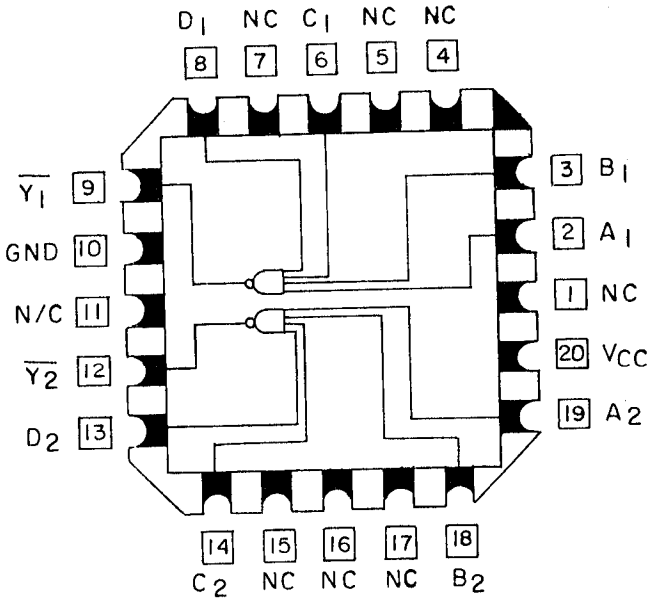


FIGURE 1. Terminal connections and logic diagram (top view) - Continued.

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Device type 01, 02

Inputs		Output
A	B	O
L	L	H
L	H	H
H	L	H
H	H	L

H = High voltage level
L = Low voltage level

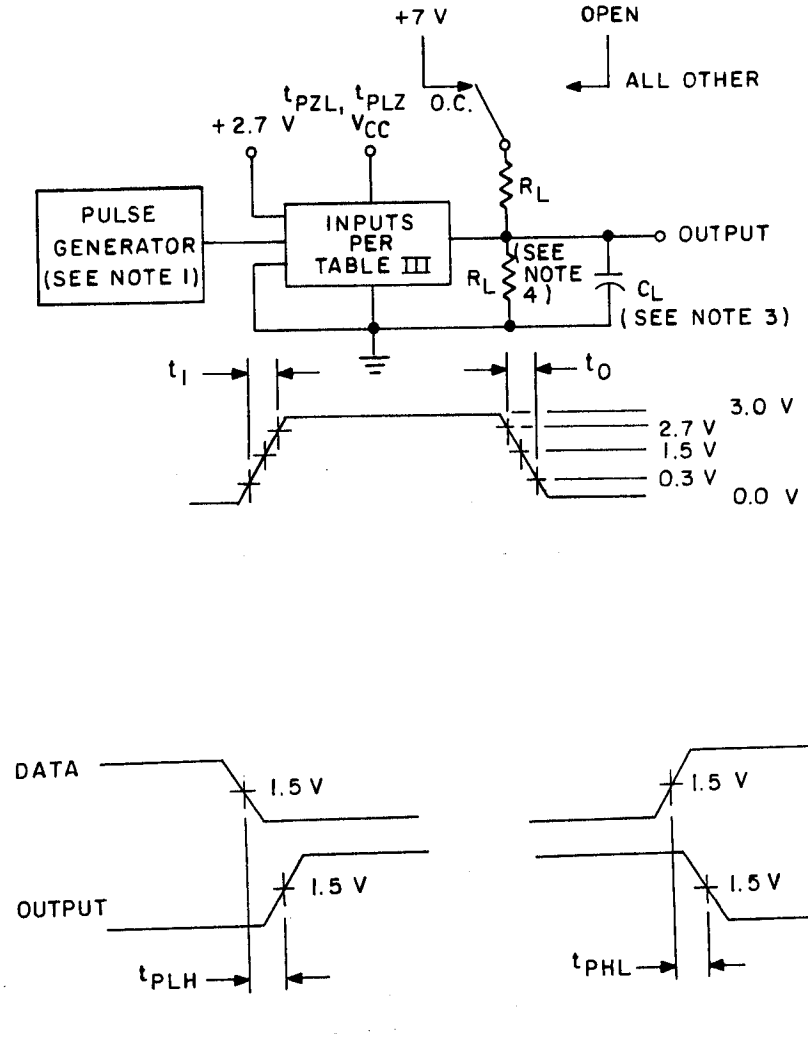
Device type 03

Inputs				Outputs
A	B	C	D	O
L	X	X	X	H
X	L	X	X	H
X	X	L	X	H
X	X	X	L	H
H	H	H	H	L

H = High voltage level
L = Low voltage level
X = Immaterial

FIGURE 2. Truth table.

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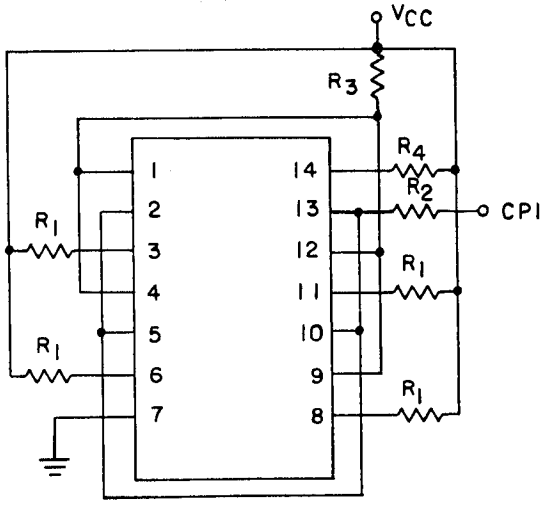


NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \cong 50\Omega$.
2. Inputs not under test are at ground or 2.7 V as specified in table III.
3. C_L = 50 pF ±10 percent, including scope probe, wiring and stray capacitance without package in test fixture.
4. RL = 499Ω ±5 percent.
5. Voltage measurements are to be made with respect to network ground terminal.

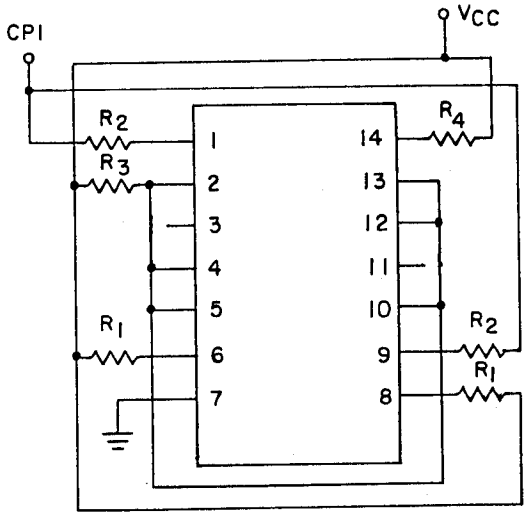
FIGURE 3. Switching test circuit and waveform for all device types.

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Device types 01 and 02



Device type 03

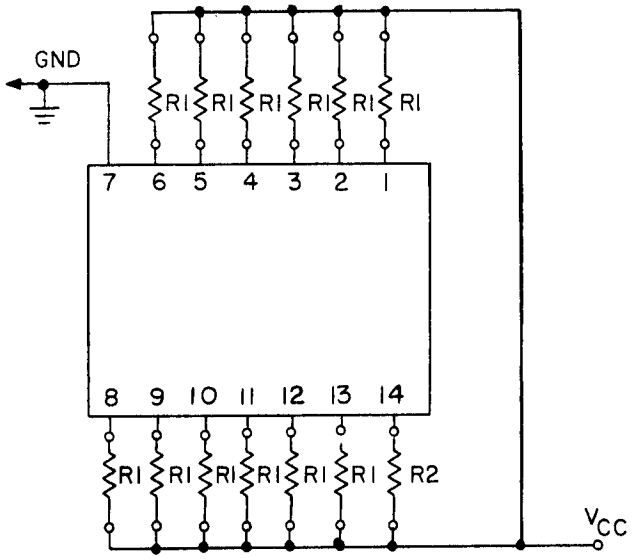


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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All device types



NOTES:
1. High temperature (175°C), condition F circuit only.
2. NC = No connection.
3. R1 = 2 kΩ
4. R2 = 2 Ω

FIGURE 4. Burn-in and life test circuits- Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open.)

Subgroup	Symbol	MIL-STD 883 method	Cases C,D Case 2 Test no.	1 2 A_1	2 3 B_1	3 4 $\overline{Y1}$	4 6 A_2	5 8 B_2	6 9 $\overline{Y2}$	7 10 GND	8 12 $\overline{Y3}$	9 13 B_3	10 14 A_3	11 16 $\overline{Y4}$	12 18 B_4	13 19 A_4	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	5.5 V	0.8 V	-1.0 mA	5.5 V	5.5 V		GND		5.5 V	5.5 V		5.5 V	5.5 V	4.5 V	$\overline{Y1}$	2.5		V
			2	0.8 V	5.5 V	-1.0 mA	"	5.5 V		"		"	"		"	"	"	$\overline{Y1}$	"		"
			3	5.5 V	"		"	0.8 V	-1.0 mA	"		"	"		"	"	"	$\overline{Y2}$	"		"
			4	"	"		0.8 V	5.5 V	-1.0 mA	"		"	"		"	"	"	$\overline{Y2}$	"		"
			5	"	"		5.5 V	"		"	-1.0 mA	"	0.8 V		"	"	"	$\overline{Y3}$	"		"
			6	"	"		"	"		"	-1.0 mA	0.8 V	5.5 V		"	"	"	$\overline{Y3}$	"		"
			7	"	"		"	"		"		5.5 V	"	-1.0 mA	"	0.8 V	"	$\overline{Y4}$	"		"
			8	"	"		"	"		"		5.5 V	"	-1.0 mA	0.8 V	5.5 V	"	$\overline{Y4}$	"		"
	V_{OL}	3007	9	"	2.0 V	48 mA	"	"		"		"	"		5.5 V	"	"	$\overline{Y1}$		0.55	"
			10	2.0 V	5.5 V	48 mA	"	"		"		"	"		"	"	"	$\overline{Y1}$		"	"
			11	5.5 V	"		"	2.0 V	48 mA	"		"	"		"	"	"	$\overline{Y2}$		"	"
			12	"	"		2.0 V	5.5 V	48 mA	"		"	"		"	"	"	$\overline{Y2}$		"	"
			13	"	"		5.5 V	"		"	48 mA	"	2.0 V		"	"	"	$\overline{Y3}$		"	"
			14	"	"		"	"		"	48 mA	2.0 V	5.5 V		"	"	"	$\overline{Y3}$		"	"
			15	"	"		"	"		"		5.5 V	"	48 mA	"	2.0 V	"	$\overline{Y4}$		"	"
			16	"	"		"	"		"		5.5 V	"	48 mA	2.0 V	5.5 V	"	$\overline{Y4}$		"	"
	V_{IC}		17	-18 mA	"		"	"		"		"	"		5.5 V	"	"	A_1		-1.2	"
			18	5.5 V	-18 mA		"	"		"		"	"		"	"	"	B_1		"	"
			19	"	5.5 V		-18 mA	"		"		"	"		"	"	"	A_2		"	"
			20	"	"		5.5 V	-18 mA		"		"	"		"	"	"	B_2		"	"
			21	"	"		"	5.5 V		"		-18 mA	"		"	"	"	B_3		"	"
			22	"	"		"	"		"		5.5 V	-18 mA		"	"	"	A_3		"	"
			23	"	"		"	"		"		"	5.5 V		-18 mA	"	"	B_4		"	"
			24	"	"		"	"		"		"	5.5 V		5.5 V	-18 mA	"	A_4		"	"
	I_{IH1}	3010	25	2.7 V						"							5.5 V	A_1		20	µA
			26		2.7 V					"							"	B_1		"	"
			27				2.7 V			"							"	A_2		"	"
			28					2. 7V		"							"	B_2		"	"
			29							"		2.7 V					"	B_3		"	"
			30							"			2.7 V				"	A_3		"	"
			31							"					2.7 V		"	B_4		"	"
			32							"						2.7 V	"	A_4		"	"
	I_{IH2}	3010	33	7.0 V						"							"	A_1		100	"
			34		7.0 V					"							"	B_1		"	"
			35				7.0 V			"							"	A_2		"	"
			36					7.0 V		"							"	B_2		"	"
			37							"		7.0 V					"	B_3		"	"
			38							"			7.0 V				"	A_3		"	"
			39							"					7.0 V		"	B_4		"	"
			40							"						7.0 V	"	A_4		"	"
	I_{IL}	3009	41	0.5 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	"	A_1	1/	1/	mA
			42	5.5 V	0.5 V		5.5 V	"		"		"	"		"	"	"	B_1	"	"	"
			43	"	5.5 V		0.5 V	"		"		"	"		"	"	"	A_2	"	"	"
			44	"	"		5.5 V	0.5 V		"		"	"		"	"	"	B_2	"	"	"
			45	"	"		"	5.5 V		"		0.5 V	"		"	"	"	B_3	"	"	"
			46	"	"		"	"		"		5.5 V	0.5 V		"	"	"	A_3	"	"	"
			47	"	"		"	"		"		"	5.5 V		0.5 V	"	"	B_4	"	"	"
			48	"	"		"	"		"		"	5.5 V		5.5 V	0.5 V	"	A_4	"	"	"
	I_{OS}	3011	49	0.0 V	0.0 V	0.0 V	"	"		"		"	"		"	5.5 V	"	$\overline{Y1}$	-100	-255	"
			50	5.5 V	5.5 V		0.0 V	0.0 V	0.0 V	"		0.0 V	0.0 V		"	"	"	$\overline{Y2}$	"	"	"
			51	"	"		5.5 V	5.5 V		"	0.0 V	0.0 V	0.0 V		"	"	"	$\overline{Y3}$	"	"	"
			52	"	"		5.5 V	5.5 V		"		5.5 V	5.5 V	0.0 V	0.0 V	0.0 V	"	$\overline{Y4}$	"	"	"
	I_{CCL} I_{CCH}	3005	53	"	"		"	"		"		5.5 V	5.5 V		5.5 V	5.5 V	"	V_{CC}		33	"
			54	0.0 V	0.0 V		0.0 V	0.0 V		"		0.0 V	0.0 V		0.0 V	0.0 V	"	V_{CC}		6	"

See footnotes at end of device type 03.

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2	Same tests and terminal conditions as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
7 T_C = +25°C	Func- tional 3/	3014	55 56 57 58	B A B A	B B A A	H " " L	B A B A	B B A A	H " " L	GND " "	H " " L	B B A A	B A B A	H " " L	B B A A	B A B A	2/ " "	All outputs "			
9 T_C = +25°C	t_{PLH}	3003	59 60 61 62 63 64 65 66	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	" " " " " " " "	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	5.0 V " " " " " " "	A_1 to $\overline{Y1}$ B_1 to $\overline{Y1}$ A_2 to $\overline{Y2}$ B_2 to $\overline{Y2}$ B_3 to $\overline{Y3}$ A_3 to $\overline{Y3}$ B_4 to $\overline{Y4}$ A_4 to $\overline{Y4}$	1.5 " " " " " " "	5.5 " " " " " " "	ns " " " " " " "
	t_{PHL}		67 68 69 70 71 72 73 74	IN 2.7 V	2.7V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	" " " " " " " "	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	" " " " " " " "	A_1 to $\overline{Y1}$ B_1 to $\overline{Y1}$ A_2 to $\overline{Y2}$ B_2 to $\overline{Y2}$ B_3 to $\overline{Y3}$ A_3 to $\overline{Y3}$ B_4 to $\overline{Y4}$ A_4 to $\overline{Y4}$	1.0 " " " " " " "	4.5 " " " " " " "	" " " " " " " "
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																					
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open.)

Subgroup	Symbol	MIL-STD 883 method	Cases C,D Case 2	1 2	2 3	3 4	4 6	5 8	6 9	7 10	8 12	9 13	10 14	11 16	12 18	13 19	14 20	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	A_1	B_1	$\overline{Y1}$	A_2	B_2	$\overline{Y2}$	GND	$\overline{Y3}$	B_3	A_3	$\overline{Y4}$	B_4	A_4	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	5.5 V	2.0 V	48 mA	5.5 V	5.5 V		GND		5.5 V	5.5 V		5.5V	5.5 V	4.5 V	$\overline{Y1}$		0.55	V
			2	2.0 V	5.5 V	48 mA	"	5.5 V		"		"	"		"	"	"	$\overline{Y1}$		"	"
			3	5.5 V	"		"	2.0 V	48 mA	"		"	"		"	"	"	$\overline{Y2}$		"	"
			4	"	"		2.0 V	5.5 V	48 mA	"		"	"		"	"	"	$\overline{Y2}$		"	"
			5	"	"		5.5 V	"		"	48 mA	"	2.0 V		"	"	"	$\overline{Y3}$		"	"
			6	"	"		"	"		"	48 mA	2.0 V	5.5 V		"	"	"	$\overline{Y3}$		"	"
			7	"	"		"	"		"		5.5 V	"	48 mA	"	2.0 V	"	$\overline{Y4}$		"	"
			8	"	"		"	"		"		5.5 V	"	48 mA	2.0 V	5.5 V	"	$\overline{Y4}$		"	"
	V_{IC}		9	-18 mA	"		"	"		"		"	"		5.5 V	"	"	A_1		-1.2	"
			10	5.5 V	-18 mA		"	"		"		"	"		"	"	"	B_1		"	"
			11	"	5.5 V		-18 mA	"		"		"	"		"	"	"	A_2		"	"
			12	"	"		5.5 V	-18 mA		"		"	"		"	"	"	B_2		"	"
			13	"	"		"	5.5 V		"		-18 mA	"		"	"	"	B_3		"	"
			14	"	"		"	"		"		5.5 V	-18 mA		"	"	"	A_3		"	"
			15	"	"		"	"		"		"	5.5 V		-18 mA	"	"	B_4		"	"
			16	"	"		"	"		"		"	5.5 V		5.5 V	-18 mA	"	A_4		"	"
	I_{IH1}	3010	17	2.7 V						"							5.5 V	A_1		20	µA
			18		2.7 V					"							"	B_1		"	"
			19				2.7 V			"							"	A_2		"	"
			20					2.7 V		"							"	B_2		"	"
			21							"		2.7 V					"	B_3		"	"
			22							"			2.7 V				"	A_3		"	"
			23							"					2.7 V		"	B_4		"	"
			24							"						2.7 V	"	A_4		"	"
	I_{IH2}		25	7.0 V						"							"	A_1		100	"
			26		7.0 V					"							"	B_1		"	"
			27				7.0 V			"							"	A_2		"	"
			28					7.0 V		"							"	B_2		"	"
			29							"		7.0 V					"	B_3		"	"
			30							"			7.0 V				"	A_3		"	"
			31							"					7.0 V		"	B_4		"	"
			32							"						7.0 V	"	A_4		"	"
	I_{IL}	3009	33	0.5 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	A_1	1/	1/	mA
			34	5.5 V	0.5 V		5.5 V	"		"		"	"		"	"	"	B_1	"	"	"
			35	"	5.5 V		0.5 V	"		"		"	"		"	"	"	A_2	"	"	"
			36	"	"		5.5 V	0.5 V		"		"	"		"	"	"	B_2	"	"	"
			37	"	"		"	5.5 V		"		0.5 V	"		"	"	"	B_3	"	"	"
			38	"	"		"	"		"		5.5 V	0.5 V		"	"	"	A_3	"	"	"
			39	"	"		"	"		"		"	5.5 V		0.5 V	"	"	B_4	"	"	"
			40	"	"		"	"		"		"	"		5.5 V	0.5 V	"	A_4	"	"	"
	I_{CEX}		41	"	0.8 V	4.5 V	"	"		"		"	"		"	5.5 V	4.5 V	$\overline{Y1}$		250	µA
			42	0.8 V	5.5 V	4.5 V	"	"		"		"	"		"	"	"	$\overline{Y1}$		"	"
			43	5.5 V	"		0.8 V	5.5 V	4.5 V	"		"	"		"	"	"	$\overline{Y2}$		"	"
			44	"	"		5.5 V	0.8 V	4.5 V	"		"	"		"	"	"	$\overline{Y2}$		"	"
			45	"	"		"	5.5 V		"	4.5 V	"	0.8 V		"	"	"	$\overline{Y3}$		"	"
			46	"	"		"	"		"	4.5 V	0.8 V	5.5 V		"	"	"	$\overline{Y3}$		"	"
			47	"	"		"	"		"		5.5 V	"	4.5 V	"	0.8 V	"	$\overline{Y4}$		"	"
			48	"	"		"	"		"		"	"	4.5 V	0.8 V	5.5 V	"	$\overline{Y4}$		"	"
	I_{CCL}	3005	49	"	"		"	"		"		"	"		5.5 V	5.5 V	5.5 V	V_{CC}		30	mA
	I_{CCH}		50	0.0 V	0.0 V		0.0 V	0.0 V		"		0.0 V	0.0 V		0.0 V	0.0 V	5.5 V	V_{CC}		7	mA

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open.)

Subgroup	Symbol	MIL-STD 883 method	Cases C,D Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Test no.	A_1	B_1	$\overline{Y1}$	A_2	B_2	$\overline{Y2}$	GND	$\overline{Y3}$	B_3	A_3	$\overline{Y4}$	B_4	A_4	V_{CC}		Min	Max	
2	Same tests and terminal conditions as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional 3/	3014	51 52 53 54	B A B A	B B A A	H " " L	B A B A	B B A A	H " " L	GND " " "	H " " L	B B A A	B A B A	H " " L	B B A A	B A B A	2/ " " "	All outputs " "			
9 T_C = +25°C	t_{PLH}	3003	55 56 57 58 59 60 61 62	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	" " " " " " " "	OUT OUT	2.7V IN	IN 2.7 V	OUT OUT	2.7 V IN	IN 2.7 V	5.0 V " " " " " " "	A_1 to $\overline{Y1}$ B_1 to $\overline{Y1}$ A_2 to $\overline{Y2}$ B_2 to $\overline{Y2}$ A_3 to $\overline{Y3}$ B_3 to $\overline{Y3}$ A_4 to $\overline{Y4}$ B_4 to $\overline{Y4}$	6.5 " " " " " " "	12.5 " " " " " " "	ns " " " " " " "
	t_{PHL}		63 64 65 66 67 68 69 70	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	" " " " " " " "	OUT OUT	2.7 V IN	IN 2.7 V	OUT OUT	2.7 V IN	IN 2.7 V	" " " " " " " "	A_1 to $\overline{Y1}$ B_1 to $\overline{Y1}$ A_2 to $\overline{Y2}$ B_2 to $\overline{Y2}$ A_3 to $\overline{Y3}$ B_3 to $\overline{Y3}$ A_4 to $\overline{Y4}$ B_4 to $\overline{Y4}$	1.0 " " " " " " "	5.0 " " " " " " "	" " " " " " " "
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																				
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open.)

Subgroup	Symbol	MIL-STD 883 method	Cases C,D / Case 2 / Test no.	1 / 2 / A_1	2 / 3 / B_1	3 / 4 / NC	4 / 6 / C_1	5 / 8 / D_1	6 / 9 / $\overline{Y1}$	7 / 10 / GND	8 / 12 / $\overline{Y2}$	9 / 13 / D_2	10 / 14 / C_2	11 / 16 / NC	12 / 18 / B_2	13 / 19 / A_2	14 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	5.5 V		5.5 V	5.5 V	-1.0 mA	GND		5.5 V	5.5 V		5.5 V	5.5 V	4.5 V	$\overline{Y1}$	2.5		V
			2	5.5 V	0.8 V		5.5 V	"	"	"		"	"		"	"	"	$\overline{Y1}$	"		"
			3	"	5.5 V		0.8 V	"	"	"		"	"		"	"	"	$\overline{Y1}$	"		"
			4	"	"		5.5 V	0.8 V	"	"		"	"		"	"	"	$\overline{Y1}$	"		"
			5	"	"		"	5.5 V		"	-1.0 mA	0.8 V	"		"	"	"	$\overline{Y2}$	"		"
			6	"	"		"	"		"	"	5.5 V	0.8 V		"	"	"	$\overline{Y2}$	"		"
			7	"	"		"	"		"	"	"	5.5 V		0.8 V	"	"	$\overline{Y2}$	"		"
			8	"	"		"	"		"	"	"	"		5.5 V	0.8 V	"	$\overline{Y2}$	"		"
	V_{OL}	3007	9	2.0 V	"		"	"	48 mA	"		"	"		"	5.5 V	"	Y1		0.55	"
			10	5.5 V	2.0 V		"	"	"	"		"	"		"	"	"	$\overline{Y1}$		"	"
			11	"	5.5 V		2.0 V	"	"	"		"	"		"	"	"	$\overline{Y1}$		"	"
			12	"	"		5.5 V	2.0 V	"	"		"	"		"	"	"	$\overline{Y1}$		"	"
			13	"	"		"	5.5 V		"	48 mA	2.0 V	"		"	"	"	$\overline{Y2}$		"	"
			14	"	"		"	"		"	"	5.5 V	2.0 V		"	"	"	$\overline{Y2}$		"	"
			15	"	"		"	"		"	"	"	5.5 V		2.0 V	"	"	$\overline{Y2}$		"	"
			16	"	"		"	"		"	"	"	"		5.5 V	2.0 V	"	$\overline{Y2}$		"	"
	V_{IC}		17	-18 mA	"		"	"		"		"	"		"	5.5 V	"	A_1		-1.2	"
			18	5.5 V	-18 mA		"	"		"		"	"		"	"	"	B_1		"	"
			19	"	5.5 V		-18 mA	"		"		"	"		"	"	"	C_1		"	"
			20	"	"		5.5 V	-18 mA		"		"	"		"	"	"	D_1		"	"
			21	"	"		"	5.5 V		"		-18 mA	"		"	"	"	D_2		"	"
			22	"	"		"	"		"		5.5 V	-18 mA		"	"	"	C_2		"	"
			23	"	"		"	"		"		"	5.5 V		-18 mA	"	"	B_2		"	"
			24	"	"		"	"		"		"	5.5 V		5.5 V	-18mA	"	A_2		"	"
	I_{IH1}	3010	25	2.7 V						"							5.5 V	A_1		20	µA
			26		2.7 V					"							"	B_1		"	"
			27				2.7 V			"							"	C_1		"	"
			28					2.7 V		"							"	D_1		"	"
			29							"		2.7 V					"	D_2		"	"
			30							"			2.7 V				"	C_2		"	"
			31							"					2.7 V		"	B_2		"	"
			32							"						2.7 V	"	A_2		"	"
	I_{IH2}		33	7.0 V						"							5.5 V	A_1		100	"
			34		7.0 V					"							"	B_1		"	"
			35				7.0 V			"							"	C_1		"	"
			36					7.0 V		"							"	D_1		"	"
			37							"		7.0 V					"	D_2		"	"
			38							"			7.0 V				"	C_2		"	"
			39							"					7.0 V		"	B_2		"	"
			40							"						7.0 V	"	A_2		"	"
	I_{IL}	3009	41	0.5 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	"	A_1	1/	1/	mA
			42	5.5 V	0.5 V		5.5 V	"		"		"	"		"	"	"	B_1	"	"	"
			43	"	5.5 V		0.5 V	"		"		"	"		"	"	"	C_1	"	"	"
			44	"	"		5.5 V	0.5 V		"		"	"		"	"	"	D_1	"	"	"
			45	"	"		"	5.5 V		"		0.5 V	"		"	"	"	D_2	"	"	"
			46	"	"		"	"		"		5.5 V	0.5 V		"	"	"	C_2	"	"	"
			47	"	"		"	"		"		"	5.5 V		0.5 V	"	"	B_2	"	"	"
			48	"	"		"	"		"		"	"		5.5 V	0.5 V	"	A_2	"	"	"
	I_{OS}	3011	49	0.0 V	0.0 V		0.0 V	0.0 V	0.0 V	"		"	"		5.5 V	5.5 V	"	$\overline{Y1}$	-100	-225	"
			50	5.5 V	5.5 V		5.5 V	5.5 V		"	0.0 V	0.0 V	0.0 V		0.0 V	0.0 V	"	$\overline{Y2}$	-100	-225	"
	I_{CCL} I_{CCH}	3005	51	5.5 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	"	V_{CC}		17	"
			52	0.0 V	0.0 V		0.0 V	0.0 V		"		0.0 V	0.0 V		0.0 V	0.0 V	"	V_{CC}		4	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open.)

Subgroup	Symbol	MIL-STD 883 method	Cases C,D Case 2 Test no.	1 2 A_1	2 3 B_1	3 4 NC	4 6 C_1	5 8 D_1	6 9 $\overline{Y1}$	7 10 GND	8 12 $\overline{Y2}$	9 13 D_2	10 14 C_2	11 16 NC	12 18 B_2	13 19 A_2	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests and terminal conditions as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional 3/	3014	53	B	B		B	B	H	GND	H	B	B		B	B	2/	All outputs			
			54	A	B		"	"	"	"	"	"	"		B	A	"	"			
			55	B	A		"	"	"	"	"	"	"		A	B	"	"			
			56	A	A		"	"	"	"	"	"	A		A	A	"	"			
			57	B	B		A	"	"	"	"	"	"		B	B	"	"			
			58	A	B		"	"	"	"	"	"	"		B	A	"	"			
			59	B	A		"	"	"	"	"	"	"		A	B	"	"			
			60	A	A		"	"	"	"	"	"	"		A	A	"	"			
			61	B	B		B	A	"	"	"	A	B		B	B	"	"			
			62	A	B		"	"	"	"	"	"	"		A	B	"	"			
			63	B	A		"	"	"	"	"	"	"		B	A	"	"			
			64	A	A		"	"	"	"	"	"	"		A	B	"	"			
			65	B	B		A	"	"	"	"	"	A		B	A	"	"			
			66	A	B		"	"	"	"	"	"	"		A	B	"	"			
			67	B	A		"	"	"	"	"	"	"		B	A	"	"			
			68	A	A		"	"	L	"	L	"	"		A	A	"	"			
9 T_C = +25°C	t_{PLH}	3003	69	IN	2.7 V		2.7 V	2.7 V	OUT	"							5.0 V	A_1 to $\overline{Y1}$	1.0	5.5	ns
			70	2.7 V	IN		2.7 V	"	"	"							"	B_1 to $\overline{Y1}$	"	"	"
			71	"	2.7 V		IN	"	"	"							"	C_1 to $\overline{Y1}$	"	"	"
			72	"	2.7 V		2.7 V	IN	"	"							"	D_1 to $\overline{Y1}$	"	"	"
			73							"	OUT	IN	2.7 V		2.7 V	2.7 V	"	D_2 to $\overline{Y2}$	"	"	"
			74							"	"	2.7 V	IN		2.7 V	"	"	C_2 to $\overline{Y2}$	"	"	"
			75							"	"	"	2.7 V		IN	"	"	B_2 to $\overline{Y2}$	"	"	"
			76							"	"	"	2.7 V		2.7 V	IN	"	A_2 to $\overline{Y2}$	"	"	"
	t_{PHL}		77	IN	2.7 V		2.7 V	2.7 V	OUT	"							"	A_1 to $\overline{Y1}$	1.0	4.5	"
			78	2.7 V	IN		2.7 V	"	"	"							"	B_1 to $\overline{Y1}$	"	"	"
			79	"	2.7 V		IN	"	"	"							"	C_1 to $\overline{Y1}$	"	"	"
			80	"	2.7 V		2.7 V	IN	"	"							"	D_1 to $\overline{Y1}$	"	"	"
			81							"	OUT	IN	2.7 V		2.7 V	2.7 V	"	D_2 to $\overline{Y2}$	"	"	"
			82							"	"	2.7 V	IN		2.7 V	"	"	C_2 to $\overline{Y2}$	"	"	"
			83							"	"	"	2.7 V		IN	"	"	B_2 to $\overline{Y2}$	"	"	"
			84							"	"	"	2.7 V		2.7 V	IN	"	A_2 to $\overline{Y2}$	"	"	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																				
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ I_{IL} limits mA minimum/maximum values for circuits as shown:

Parameter	Circuit A	Circuit B	Circuit C	Circuit D
I_{IL}	-.50/-1.2		0/-1.2	

2/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

3/ A = 2.5 V minimum; B = 0.55 V or GND, H ≥ 2.5 V, L ≤ 0.5V.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, E, or F, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5005 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F37
02	54F38
03	54F40

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6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit				
	A Fairchild Semiconductor	B Motorola	C Signetics	D Texas Instrument	E
01	X		X		
02	X		X		
03	X				

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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